

Multiple Choice

1. **Answer:** B

Options: A) Magnetic Resonance Imaging; B) Dual-energy X-ray absorptiometry; C) Isotope dilution; D) Skinfold Thickness

Note: Dual-energy X-ray absorptiometry is the correct answer because it uses low-level X-rays to differentiate between bone mineral content and soft tissue, providing precise measurements of body mineral content.

2. **Answer:** C

Options: A) <0.8 g protein per kg bodyweight per day; B) 0.8-1.2 g protein per kg bodyweight per day; C) 1.2-1.7 g protein per kg bodyweight per day; D) >2.0 g protein per kg bodyweight per day

Note: Athletes require 1.2-1.7 grams of protein per kilogram of body weight daily to support muscle repair and growth, which is essential for their training and performance.

3. **Answer:** D

Options: A) Decreased calcium excretion in the urine; B) Increased risk of fracture; C) Decreased dietary calcium absorption; D) Negative calcium balance and bone mineral loss

Note: Higher dietary sodium intake can lead to increased calcium excretion in urine, resulting in negative calcium balance and potential bone mineral loss.

4. **Answer:** A

Options: A) When intakes exceed established ULs (Upper Levels); B) At any level greater than a recommended dietary intake; C) Only when intakes are less than those required for nutritional benefits; D) Under no known conditions of usual intake

Note: Nutrients can pose health risks when consumed in excessive amounts, as established Upper Levels (ULs) indicate the maximum daily intake that is unlikely to cause adverse health effects.

5. **Answer:** C

Options: A) The role of iron in myelination; B) The role of iron in neurochemistry; C) The role of iron in dopamine production; D) None of the above

Note: The correct answer is based on the fact that iron is essential for the synthesis of dopamine, a neurotransmitter that plays a crucial role in regulating mood, motivation, and emotional responses, which directly influences socio-emotional development such as shyness and engagement.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The most important function of the gastrointestinal tract right after eating is the digestion and absorption of nutrients, rather than the elimination of toxins.

7. **Answer:** True

Options: A) True; B) False

Note: The absorption of glucose in the gastrointestinal tract is linked to the uptake of sodium because glucose is co-transported with sodium ions across the intestinal cell membranes through a mechanism called secondary active transport, which relies on the sodium gradient established by the sodium-potassium pump.

8. **Answer:** False

Options: A) True; B) False

Note: High protein diets can promote weight loss by increasing feelings of fullness and helping to preserve lean muscle mass during calorie restriction.

9. **Answer:** True

Options: A) True; B) False

Note: Type IIb muscle fibers are primarily designed for explosive movements and rely on anaerobic metabolism, which results in low oxidative capacity and a tendency to fatigue quickly.

10. **Answer:** True

Options: A) True; B) False

Note: Bariatric surgery leads to significant weight loss and metabolic changes that improve insulin sensitivity and glucose metabolism, thereby reducing the incidence of type 2 diabetes in obese individuals by 80-90%.

Long Form Response

11. **Answer:** The Atherogenic Lipoprotein Phenotype (ALP) is characterized by three main abnormalities: reduced levels of high-density lipoprotein (HDL), a high proportion of small dense low-density lipoprotein (LDL), and elevated triglycerides. HDL is known as "good" cholesterol because it helps remove cholesterol from the bloodstream, so lower levels can lead to increased plaque buildup in arteries. Small dense LDL particles are particularly harmful as they can more easily penetrate the arterial walls and contribute to atherosclerosis, a condition that narrows and hardens the arteries. Elevated triglycerides also add to the risk of cardiovascular diseases. Together, these lipoprotein abnormalities can significantly increase the risk of heart disease, stroke, and other health issues, highlighting the importance of maintaining healthy lipid profiles for overall health.

Note: correct because it identifies the key features of the Atherogenic Lipoprotein Phenotype (ALP), explains the role of HDL in cholesterol removal, and details how

small dense LDL and elevated triglycerides contribute to increased cardiovascular risk through arterial plaque formation.

12. **Answer:** The nutritional status of older adults is crucial for their overall health, particularly concerning iron levels. While iron deficiency can lead to anemia and fatigue, high iron stores pose even greater risks in this population. Excess iron can accumulate in the body and contribute to oxidative stress, increasing the likelihood of chronic diseases such as heart disease and diabetes. Older adults often have reduced ability to regulate iron absorption, making them more vulnerable to these risks. Therefore, maintaining balanced iron levels is essential for the health and well-being of older adults, as high iron stores can be more harmful than having too little iron.

Note: The answer correctly highlights that while iron deficiency can cause anemia, older adults are at greater risk from high iron stores due to the potential for oxidative stress and chronic diseases, compounded by their decreased ability to regulate iron absorption.